When Trackers Fail: VLM-Guided Verification and Recovery for Robust Video Object Segmentation

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Abstract

Recent trackers built on foundation models such as the Segment Anything Model (SAM) achieve strong performance by propagating object masks through memory. However, this strength is also a key source of failure: under occlusion, reappearance, or distractors, an incorrect prediction can enter the tracking loop and be propagated over time, resulting in persistent identity drift. Existing approaches seek to prevent such failures through improved memory and temporal modeling, but once a failure occurs, the tracker has no explicit mechanism to recognize or correct it. We address this gap with Dual-Agent, a failure-aware tracking framework that complements propagation with explicit verification and recovery. We address this gap with Dual-Agent, a failure-aware tracking framework that complements propagation with explicit verification and recovery. A vision-language verification agent first reasons over spatio-temporal context to determine whether the tracker is still following the target. When a failure is detected, a correction agent uses temporal context to recover the target before tracking continues. The correction agent is trained on real SAM3 failures, augmented with targeted identity-switch and distractor perturbations that expose it to challenging ambiguities during recovery. On LVOS-v2 and SecVOS, Dual-Agent improves SAM3 by 1.2 and 2.7 $J\&F$, respectively. Importantly, the proposed framework transfers without retraining to SAM2.1, SeC, and Cutie, as well as to long-term video object tracking benchmarks including LaSOT-ext, TNL2K, and VOT-LT2022. Our results show that explicitly reasoning about when tracking fails and how to recover provides a complementary path to robust tracking beyond improving propagation alone.

1 Introduction

Segmentation-based tracking has recently undergone a paradigm shift with the emergence of foundation models such as Segment Anything Model and its video extensions (e.g., SAM2 [19] and SAM3 [4]), which enable promptable mask propagation across frames. These approaches achieve strong performance by combining powerful visual representations with streaming memory, and have quickly become the dominant paradigm for video object segmentation and tracking. However, despite their impressive generalization, they remain fundamentally brittle in challenging scenarios involving occlusion, reappearance, and distractors. In such cases, trackers often drift toward visually similar objects or fail to recover the target after disappearance, revealing a lack of explicit scene understanding and failure awareness (see Figure 1).

A growing body of recent work has focused on improving SAM-based trackers by refining temporal modeling and memory design. For example, SAMURAI [27] introduces motion-aware memory selection to incorporate temporal motion cues and mitigate error propagation. Similarly, DAM4SAM [20]

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

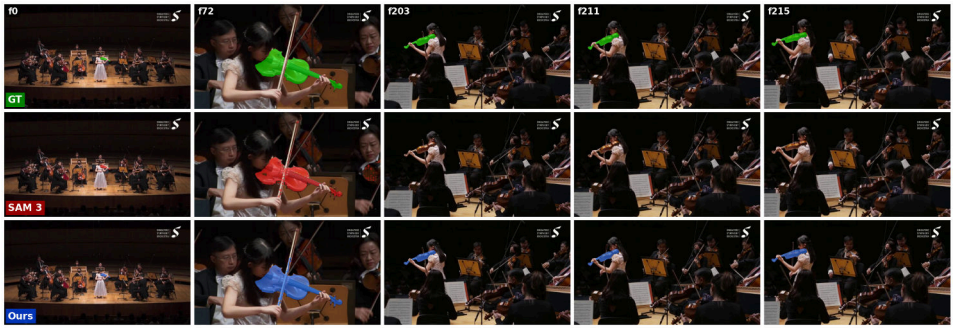


Figure 1: Our proposed dual-agent framework recovers tracking errors made by the tracker. Rows show ground truth (top), the SAM3 baseline (middle), and our dual-agent framework (bottom).

proposes a distractor-aware memory mechanism that explicitly models confusing objects and improves robustness under occlusion. Other extensions explore long-term memory and improved memory management strategies [9]. More recently, SeC [28] introduces progressive concept construction to improve robustness under complex scene changes by incorporating higher-level semantic priors into segmentation. While these approaches achieve substantial gains, they all share a common design principle: *tracking is performed via recursive propagation of previous predictions*. As a result, failures caused by occlusion, distractors, or identity ambiguity are not explicitly identified; incorrect predictions are simply propagated forward, often leading to persistent drift.

In parallel, recent advances in vision-language models (VLMs) have demonstrated strong capabilities in semantic understanding and contextual reasoning [2, 7]. Motivated by these properties, a growing line of work in referring and reasoning-based video object segmentation incorporates VLMs to identify or disambiguate target objects based on language or contextual cues [23, 8, 18]. However, in these approaches, VLMs are primarily used for object grounding or selection, while the actual tracking remains handled by a conventional propagation-based model. Thus, while they improve what to track, they do not address a critical question: *whether the tracker remains correct over time*. In particular, they lack an explicit mechanism for detecting and correcting failures during tracking.

In this work, we argue that the central limitation of modern tracking systems is not the lack of better propagation, but *the absence of failure-aware reasoning*. Existing approaches implicitly assume that predictions remain valid over time, and thus focus on improving temporal consistency. However, as evidenced by recent SAM-based extensions, even sophisticated memory and motion modeling cannot fully prevent drift once failures occur, since they are never explicitly diagnosed or corrected [27, 20].

To address this, we propose a fundamentally different formulation that introduces explicit reasoning into the tracking loop. Rather than relying on propagation alone, our key idea is to complement it with two capabilities: (i) verifying whether the current prediction remains correct, and (ii) recovering the target when it does not. We instantiate this idea with *Dual-Agent*, a framework comprising a verification agent and a correction agent alongside the underlying propagator. The verification agent is a vision-language reasoning model that operates over spatio-temporal context to assess whether the current prediction still corresponds to the target. Rather than relying solely on heuristic confidence scores, it reasons about object identity, visibility, and consistency across frames, enabling it to recognize identity switches, missed detections, and false positives during occlusion and reappearance.

When a failure is detected, the correction agent is selectively activated to recover the target using temporal context before propagation continues. Effective recovery, however, requires resolving precisely the identity ambiguities that cause tracking to fail. We therefore train the correction agent on failures mined from SAM3 and augment them with targeted identity-switch and distractor-centric perturbations. These perturbations expose the model to challenging target–distractor ambiguities and encourage it to reason about target identity across time rather than relying on local appearance alone. The separation of verification from correction allows the framework to intervene specifically when the propagated prediction becomes unreliable, while otherwise retaining the underlying tracker.

We evaluate Dual-Agent across video object segmentation and long-term object tracking benchmarks. On LVOSv2 and SecVOS, Dual-Agent improves SAM3 by 1.2 and 2.7 $J\&F$, respectively, while on LaSOT-ext, TNL2K, and VOT-LT2022, it yields gains of 1-2% in long-term object tracking.

Beyond SAM3, the proposed framework transfers without retraining to other propagators, including SAM2.1, SeC, and Cutie, demonstrating that the verification and recovery mechanisms generalize across different propagation-based trackers.

Overall, our Dual-Agent framework with SAM3 as the propagator reframes segmentation-based tracking from improving propagation to explicitly modeling failure detection and recovery. By integrating structured reasoning into the tracking loop, our approach addresses a fundamental limitation shared by both SAM-based trackers and prior VLM-assisted methods. More broadly, our results suggest that incorporating reasoning into perception systems is a promising direction for building robust, self-correcting vision models capable of operating in complex, real-world environments.

2 Related Work

Memory-based video object segmentation. Modern semi-supervised VOS systems are dominated by memory-based propagation. XMem [5] established the importance of explicit long-term and short-term memory for stable mask propagation, while later methods such as Cutie [6] improved object-centric memory reading and multi-object robustness. Related work has further explored stronger retrieval and identity modeling through object-centric matching and retrieval mechanisms [21, 25], as well as unified prompt-driven frameworks such as TarViS [1] and UniVS [17]. Foundation-model-based trackers including SAM2 [19], SAM2Long [9], SAMURAI [27], DAM4SAM [20], and SAM3 [4] further improve propagation through motion-aware memory selection, distractor-aware memory writing, and long-video memory management. However, these approaches still rely on recursive propagation of previous predictions and do not explicitly reason about whether the tracker has already failed. Once drift occurs, corrupted state can be propagated and reinforced over time. In contrast, our approach introduces an explicit reliability layer on top of propagation that detects and selectively corrects failures before they dominate future tracking.

VLMs for video understanding and tracking. VLMs have recently demonstrated strong capabilities in semantic grounding, contextual disambiguation, and visual reasoning. Language-guided VOS methods such as ReferFormer [23], RefSAM [18], and SAMWISE [8] use language prompts to specify target objects and adapt the propagation pipeline accordingly. More recent reasoning-oriented approaches including VISA [26], ViLLa [29], CoT-RVS [15], VideoLISA [3], and VideoSeg-R1 [24] incorporate higher-level semantic reasoning into video segmentation. Training-free agentic methods such as AL-Ref-SAM2 [12], AgentRVOS [14], and Refer-Agent [13] similarly combine large multimodal models with SAM-style propagation, while SeC [28] injects concept-level priors directly into the backbone to improve robustness under complex scene changes. However, these methods primarily use reasoning to improve target grounding, representation, or segmentation quality during propagation. Unlike SeC that probes and refines intermediate tracker representations, our framework does not modify the propagator internals or memory features. In contrast, we introduce explicit failure-aware reasoning into the tracking loop itself. Our framework reasons over tracker-generated, potentially corrupted temporal context to determine whether the current track remains reliable, and selectively triggers recovery when failures such as identity drift or missed reappearance are detected.

3 Methodology

3.1 Problem Formulation

Given a video $\{I_t\}_{t=1}^{T}$ and an initial mask prompt on the first frame, a tracker produces a sequence of masks $\{\hat{M}_t\}_{t=1}^{T}$. In SAM3-style propagation, the estimate at frame t is produced from the current image and an internal temporal state, namely $\hat{M}_t = \mathcal{T}(I_t, \mathcal{S}_{t-1})$, where $\mathcal{S}_{t-1}$ summarizes memory from previous frames. The next state $\mathcal{S}_t$ is then updated from the current prediction. This recursive formulation is powerful, but it creates a failure mode that is especially severe under occlusion and distractors: once the tracker predicts wrong object or loses target, the error is fed back into memory and propagated forward. Our goal is to augment this propagation with explicit failure awareness. Rather than trusting every prediction equally, we seek a mechanism that can (i) determine when the tracker is likely wrong, and (ii) selectively recover the target when failure occurs. We therefore keep SAM3 as the propagator and target the regime in which propagation becomes unreliable. This yields

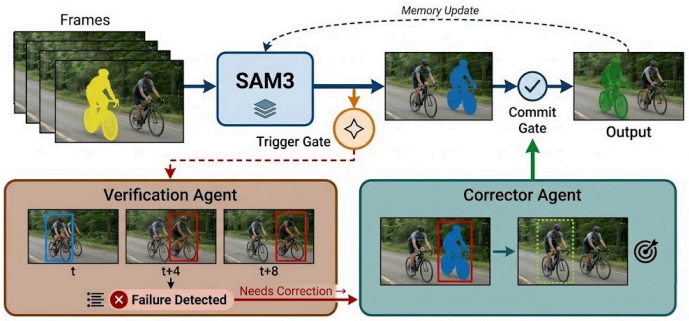


Figure 2: Overview of the proposed dual-agent framework. SAM3 performs memory-based propagation, while a sparse trigger gate selectively invokes a verification agent to assess whether the current tracker state remains reliable. When a failure is detected, a correction agent re-localizes the target, and a commit gate decides whether the corrected state should overwrite the current propagation state.

a different view of VOS from most prior work: the central problem is not only how to propagate masks better, but how to regulate propagation once its internal state becomes untrustworthy.

The resulting framework is guided by four design principles that also summarize the main technical contributions of the paper. First, we use an *asymmetric decomposition* of the problem into verification and correction, because deciding *whether* the track is wrong and deciding *where* the target is should not be treated as the same task. Second, we use a *tracker-aware reasoning interface*: the verifier reasons over a prompt that exposes both trusted exemplars and tracker-produced potentially corrupted context, rather than over clean oracle trajectories. Third, we use a *state-safe intervention policy* combining sparse triggers, and commit gating, motivated by the fact that an incorrect correction is costly because it rewrites future tracker memory. Fourth, we use *failure-oriented supervision* for recovery, including identity-switch and distractor-centric perturbations that explicitly train the corrector on the kinds of ambiguity that cause SAM-style propagators to fail.

3.2 Dual-Agent Framework

We formulate our approach as a dual-agent framework that uses SAM3 as propagator and adds two auxiliary agents on top: a verification agent $\mathcal{A}$ and a correction agent $\mathcal{C}$ (Figure 2). The verifier $\mathcal{A}$ decides whether the current tracked hypothesis is trustworthy; the corrector $\mathcal{C}$ is invoked only when $\mathcal{A}$ predicts failure. The dual-agent decomposition keeps these objectives separate and turns reasoning into an explicit control layer over propagation rather than an internal modification of the propagator.

Let I_t denote the video frame at time t, $\mathcal{S}_t$ the internal SAM3 memory state after processing frame t, and $\hat{M}_t = \mathcal{T}(I_t, \mathcal{S}_{t-1})$ the propagated mask at time t. We convert $\hat{M}_t$ to a box hypothesis $\hat{b}_t = \mathcal{B}(\hat{M}_t)$ and denote the SAM3 confidence score by $s_t \in [0, 1]$. The controller is evaluated only at a sparse trigger set $\mathcal{Q} \subset \{1, \ldots, T\}$. At each $t \in \mathcal{Q}$, we build a visual prompt $\mathcal{P}_t$ from tracker history and query frame I_t, and apply $v_t = \mathcal{A}(\mathcal{P}_t) \in \{0, 1\}$, where $v_t = 1$ denotes `incorrect`. When $v_t = 1$, the correction agent predicts a visibility variable $\tilde{o}_t \in \{0, 1\}$ and a corrected box $\tilde{b}_t \in \mathbb{R}^4$: $(\tilde{o}_t, \tilde{b}_t) = \mathcal{C}(\mathcal{P}_t)$. The tracker state is overwritten only if a separate commit rule is satisfied. Hence the final output box $b_t^\star$ obeys

$$b_t^\star = \begin{cases} \tilde{b}_t, & \text{if } t \in \mathcal{Q},\ v_t = 1,\ \Gamma_t = 1,\ \tilde{o}_t = 1, \\ \emptyset, & \text{if } t \in \mathcal{Q},\ v_t = 1,\ \Gamma_t = 1,\ \tilde{o}_t = 0, \\ \hat{b}_t, & \text{otherwise}, \end{cases} \tag{1}$$

where Γ_t is the commit gate defined in Section 3.3. If a correction is committed, SAM3 is re-prompted at frame t and propagation resumes from the corrected state rather than from the original prediction. The system therefore intervenes through sparse state resets rather than dense post-hoc mask refinement. In SAM-style trackers, the main cost of an error is not only that one frame is wrong, but that the wrong state is propagated into future frames.

3.2.1 Verification Agent

Both agents jointly process visual frames and textual instructions.

The verifier's goal is to check if the target is being correctly tracked in the current frame given the past target trajectory. For a query time t, let $\mathcal{P}_t = \big(E_t,\, U_t,\, R_t,\, Q_t\big)$ denote the visual prompt, where E_t contains exemplar frames, U_t contains uniformly sampled past context, R_t contains recent context, and Q_t is the query frame. In the main configuration, $|E_t| = 2$, $|U_t| = 2$, and $|R_t| = 2$, yielding a 7-image prompt in total. The exemplars are the earliest visible appearances of the target. The uniform context is sampled from the prior trajectory, while the recent context is sampled with a fixed stride from the immediate past.

Each frame in $\mathcal{P}_t$ carries a bounding box overlay that exposes the tracker state to the VLM. Exemplar images show ground-truth boxes in green. Context images show tracker-produced boxes in red, or no box if the tracker predicts absence. The query frame similarly shows the current tracker hypothesis $\hat{b}_t$ in red when present. These context boxes are not oracle boxes: the verifier observes the same possibly corrupted trajectory that the deployed tracker has produced so far. Clean temporal context would define an easier problem than the one faced at inference time and would break the alignment between verification and deployment. Formally, the verifier computes $\mathcal{A}(\mathcal{P}_t) = v_t$, where v_t is the binary correctness label. We instantiate $\mathcal{A}$ with Qwen3-VL-8B-Thinking [2] and used it zero-shot with a structured assessment prompt (see Appendix D). The prompt asks the model to first locate the target instance in the query image and only then judge whether the red tracker box is on the correct object. This ordering promotes identity-level reasoning rather than local box validation, which is critical when the dominant failures are semantic swaps to plausible distractors.

To reduce single-sample variance, we optionally use repeated verifier calls on temporally shifted query points and aggregate them by majority vote. If the vote count is K, the fire decision becomes $\bar{v}_t = \mathbf{1}\Big[\sum_{k=1}^{K} v_t^{(k)} > \frac{K}{2}\Big]$. This voting mechanism is especially useful in borderline occlusion and reappearance cases, where a single reasoning trace may be unstable.

3.2.2 Correction Agent

The correction agent $\mathcal{C}$ uses the same prompt layout as $\mathcal{A}$ but solves a different task: conditional re-localization under tracker failure. Rather than asking for a binary correctness judgment, we ask the model to predict whether the target is visible in the query frame and, if so, to emit a corrected 2D box. The model outputs $\mathcal{C}(\mathcal{P}_t) = \big(\tilde{o}_t, \tilde{b}_t^{\mathrm{norm}}\big)$, where $\tilde{b}_t^{\mathrm{norm}} \in [0, 1000]^4$ follows the Qwen-VL normalized coordinate convention.

We decouple correction from verification rather than asking the same model to both detect and repair failures. The corrector can then specialize to the error distribution induced by SAM3: identity drift, missed visible targets, false positives during occlusion, and ambiguous reappearance. The verifier is optimized for precision in deciding *whether* intervention is warranted, while the corrector is optimized for accuracy in deciding *where* the target should be. In the current system, $\mathcal{C}$ is a Qwen3-VL-4B-Instruct [2] model fine-tuned for structured grounded output, which is substantially lighter than the verifier and better suited to repeated localization. The framework therefore allocates semantic reasoning, recovery, and propagation to different roles according to their function rather than forcing a single model to absorb all three.

The implementation also supports iterative correction. After a candidate correction is produced, the verifier may re-evaluate the proposed box; if it is rejected, the corrector can be called again with the rejected box as additional context. Denoting the r-th correction by $\tilde{b}_t^{(r)}$, this defines a short recovery chain $\tilde{b}_t^{(r)} = \mathcal{C}\Big(\mathcal{P}_t, \tilde{b}_t^{(r-1)}\Big)$, $r = 1, \ldots, R.$. Iterative refinement is particularly useful in challenging recovery scenarios where a single correction step may fail to fully recover.

3.3 Selective Triggering and Gating

Applying a VLM at every frame is both computationally expensive and potentially harmful, since unnecessary corrections can overwrite a valid tracker state. We therefore expose the agents through a sparse trigger process and require additional gates before the corrector can modify the tracker state.

Trigger policy. We define a sparse trigger set $\mathcal{Q}$ using three cues from SAM3: occlusion, reappearance, and instability in the tracker's own confidence stream. The first two are event-based. During occlusion, we poll every M frames, while reappearance triggers are delayed by at least N frames to avoid unstable transient predictions. The third cue is an effective-IoU-variance trigger designed to catch emerging drift before the tracker has fully failed. Recall that SAM3 produces a scalar confidence s_t from its objectness and mask-quality estimates. Rather than thresholding s_t directly, we monitor its short-horizon instability. Let $\mu_t = \frac{1}{W}\sum_{i=0}^{W-1} s_{t-i}, \qquad \sigma_t^2 = \frac{1}{W}\sum_{i=0}^{W-1}(s_{t-i}-\mu_t)^2$ denote the mean and variance of the recent confidence window. We trigger the verifier when σ_t^2 exceeds a threshold for a small number of consecutive steps. Intuitively, healthy propagation tends to produce a stable confidence trace, whereas identity swaps, distractor attraction, or partial occlusion often induce oscillatory confidence even before a hard disappearance event is declared. The variance trigger therefore acts as an early-warning signal for memory corruption, complementing the more explicit occlusion and reappearance triggers.

Trigger settings are selected on a held-out SA-V training shard, disjoint from corrector training and all evaluation benchmarks, and fixed for evaluation. Appendix J details the calibration population and the comparison with dense verification.

Commit gate. At triggered frames, we bypass the verifier when SAM3 is already confident. Specifically, if $s_t > \tau_{\text{conf}}$, we directly accept the propagated prediction. This focuses reasoning on uncertain regimes where memory corruption is most likely. When the verifier predicts failure, the corrector proposes a visibility decision $\tilde{o}_t$ and a corrected box $\tilde{b}_t$. We commit the correction only if it differs substantially from the current tracker hypothesis. When both boxes are present, we compute $d_t = \text{IoU}(\tilde{b}_t, \hat{b}_t)$, and overwrite the tracker state only if $d_t < 0.5$. This prevents unnecessary memory rewrites from near-identical corrections. Overall, the controller enters correction mode only when $s_t < \tau_{\text{conf}}, \bar{v}_t = 1, t \in \mathcal{Q}$ and commits the correction only if $\text{IoU}(\tilde{b}_t, \hat{b}_t) < 0.5$. The resulting commit gate is

$$\Gamma_t = \mathbf{1}[t \in \mathcal{Q}]\mathbf{1}[s_t < \tau_{\text{conf}}]\mathbf{1}[\bar{v}_t = 1]\mathbf{1}[d_t < 0.5], \tag{2}$$

with the final factor set to 1 when the corrector predicts absence.

3.4 Failure-Oriented Training Data

Data collection. The correction agent is trained on data mined from SAM3 propagation trajectories rather than oracle trajectories, so that supervision matches the failure distribution encountered at test time. We initialize SAM3 with the official SA-V masklet at each object's first-appearance frame and propagate to the end of the video on `sav_000`–`sav_005` [19] (5,145 sequences, 16,045 objects). For each annotated frame, we record SAM3's prediction, confidence, IoU against ground truth, and a set of visually similar distractor candidates, producing a structured per-object trace.

Sample generation. Query frames are selected with a failure-oriented schedule that oversamples frames where SAM3 disagrees with ground truth, while retaining a small fraction of high-IoU frames. For each query, we render the same seven-image prompt used at test time: two early exemplar frames, two uniformly sampled history frames, two recent-context frames, and the query frame itself. The target is a structured visibility decision and ground-truth box. To expose the corrector to identity ambiguity, we additionally apply an identity-switch perturbation with a certain probability in which one or two recent context boxes, and sometimes the query box, are replaced by mined distractor boxes from the same frame. The resulting corpus contains 79,457 samples, split sequence-stratified into 75,341 train and 4,116 validation samples. More details are in the supplementary.

3.5 Training Objective and Inference

The learned correction model is obtained by supervised fine-tuning a Qwen3-VL-4B-Instruct backbone with LoRA. Given a training sample $(\mathcal{P}_t, y_t)$, where y_t is a structured JSON answer containing visibility and, a normalized box, we optimize the standard autoregressive objective $\mathcal{L}_{\text{SFT}} = -\sum_{n=1}^{N_t} \log p_\theta(y_{t,n} \mid y_{t,<n}, \mathcal{P}_t)$. The full test-time procedure can be summarized as follows. SAM3 is first mask-prompted at first appearance and propagates frame by frame and maintains its internal memory state $\mathcal{S}_{t-1}$. At a triggered frame $t \in \mathcal{Q}$, we compute the prompt $\mathcal{P}_t$ and query the verifier unless the confidence short-circuit accepts the current prediction. If the verifier declares the

track correct, we keep $\hat{b}_t$ and continue propagation. If the verifier declares the track incorrect, we query the corrector, evaluate the commit gate, and, when the gate fires, overwrite the tracker prompt at frame t with corrected state. Equivalently, the memory update can be written as

$$\mathcal{S}_t = \begin{cases} \mathcal{U}(\mathcal{S}_{t-1}, I_t, b_t^\star), & \Gamma_t = 1, \\ \mathcal{U}\Big(\mathcal{S}_{t-1}, I_t, \hat{b}_t\Big), & \Gamma_t = 0, \end{cases} \tag{3}$$

where $\mathcal{U}$ denotes the SAM3 state update induced by the accepted prompt at frame t. The key point is that the controller does not merely replace the output on one frame; it changes the future trajectory by altering the memory that subsequent frames will read from. This is why selective triggering and conservative commitment are central to the method: one incorrect correction can pollute many future frames, while one correct reset can recover a long segment after drift.

4 Experiments

Experimental Setup. We evaluate Dual-Agent on five benchmarks covering video object segmentation (VOS) and long-term object tracking. For VOS, we use LVOS-v2 valid [11], comprising 140 sequences and 239 objects with dense per-frame annotations, and SecVOS [28], comprising 160 sequences and 182 objects with an emphasis on scene changes and multi-shot discontinuities. We follow the scene-stratified evaluation protocol of SeC for SecVOS and report the standard VOS metrics: region similarity ($\mathcal{J}$), contour accuracy ($\mathcal{F}$), and their mean ($\mathcal{J}\&\mathcal{F}$). For long-term object tracking, we evaluate on LaSOT-ext [10] (150 sequences), TNL2K [22] (700 sequences), and VOT-LT2022 [16] (50 sequences). We report AUC, precision at 20 pixels, and normalized precision at 0.2 for LaSOT-ext and TNL2K, and precision, recall, and F-score for VOT-LT2022. Further details on the tracking protocols are provided in Appendix L.

The verifier uses Qwen3-VL-8B-Thinking in a zero-shot setting, while the correction agent uses a fine-tuned Qwen3-VL-4B-Instruct. We use three verification votes with a stride of 4 and three correction rounds. SAM3 remains frozen throughout all experiments. Additional implementation details are provided in the supplementary material.

Quantitative Results. Our experiments are designed to validate three questions central to the proposed framework. First, can the correction agent recover targets in regimes where recursive propagation fails? Second, does explicit failure-aware reasoning improve a strong SAM3 propagator on challenging long-video benchmarks? Third, do these gains arise from sparse and well-targeted interventions rather than frequent rewrites of tracker state?

Table 1: Corrector-only validation on the held-out corrector-SFT split. Each sample is one query frame from the 7-frame prompt window; results report IoU against ground truth for SAM3 and the corrector on the directly measurable subset (1,888 samples).

SAM3 IoU regime	Samples	SAM3 IoU	Corrector IoU	Δ	Wins / Losses
< 0.1 (lost target)	347	0.024	0.612	+0.587	270 / 11
0.1–0.4 (severe drift)	808	0.224	0.727	+0.503	688 / 76
0.4–0.7 (drift)	182	0.492	0.734	+0.242	120 / 16
≥ 0.7 (already good)	551	0.955	0.912	-0.043	9 / 40

We first evaluate the correction agent in isolation to determine whether it can recover targets once SAM3 has already drifted or lost identity. Table 1 reports validation performance on the held-out corrector-SFT split, with one query frame sampled from each 7-frame prompt window. For each query frame, we compare the IoU of SAM3's prediction and the corrector's output against ground truth. The results show a clear asymmetric pattern. The corrector substantially improves localization once SAM3 has drifted or lost the target entirely, achieving gains of +0.587 IoU in the lost-target regime and +0.503 IoU under severe drift. In contrast, when SAM3 is already correct, the corrector provides little benefit and can slightly reduce accuracy. This validates the core motivation behind the dual-agent design: recovery is valuable precisely in the regimes where recursive propagation becomes unreliable, while unnecessary intervention must be avoided through explicit verification.

We then evaluate full closed-loop system on LVOS-v2. Table 2 places the proposed framework against prior LVOS-v2 methods and also reports SAM3 baseline. Besides the overall $\mathcal{J}\&\mathcal{F}$ score, the

Table 2: LVOS-v2 validation comparison against prior methods. $\mathcal{J}\&\mathcal{F}$ is the overall score; J_s/F_s and J_u/F_u denote seen- and unseen-object region/boundary scores.

Method	$\mathcal{J}\&\mathcal{F}$	J_s	F_s	J_u	F_u
LWL	60.6	58.0	64.3	57.2	62.9
RDE	62.2	56.7	64.1	60.8	67.2
STCN	60.6	57.2	64.0	57.5	63.8
XMem	64.5	62.6	69.1	60.6	65.6
SAM 2.1	84.1	80.7	87.4	80.6	87.7
SAM2.1Long	85.9	81.7	88.6	83.0	90.5
SeC	86.5	—	—	—	—
SAM3	88.2	**85.4**	**92.9**	83.3	91.5
Dual-agent (ours)	**89.4**	84.9	92.7	**85.8**	**94.1**

Table 3: Mechanism-level analysis on SecVOS: (a) direct attribution by SAM3 state at commit time, and (b) controller activity. In (b), `Verifier`, `Corrector`, and `Commit` count verifier calls, corrector calls, and committed corrections; `Outcome` reports better / worse / neutral box-IoU changes relative to SAM3 at the 52 commits with a ground-truth box; the other 36 commits are excluded.

(a) Direct Attribution

State at commit	Count	SAM3 IoU	Dual IoU
`sam3_missed`	39	0.000	0.856
`sam3_drifted`	4	0.283	0.476
`sam3_was_ok`	9	0.926	0.049
`no_gt_at_commit`	36	—	—

(b) Controller Statistics

Trigger	Verifier	Corrector	Commit	Outcome
Overall	1360	240	88	40/11/1
Eff IoU var	713	120	53	23/6/1
Occlusion	508	105	26	17/2/0
Reappear	139	15	9	0/3/0

table includes seen-object and unseen-object breakdowns through J_s/F_s and J_u/F_u. The proposed framework improves SAM3 from 88.2 to 89.4 $\mathcal{J}\&\mathcal{F}$ while achieving the strongest unseen-object performance in the table. These results demonstrate that explicit failure-aware reasoning remains beneficial even on top of a strong propagator without modifying the propagator itself. The slight drop on seen subset might be because the verifier can occasionally overrule correct SAM3 predictions.

SecVOS provides the clearest test of the paper's central claim. The benchmark is built around scene discontinuity, reappearance, and abrupt semantic change, so it directly probes the regime in which propagation can remain locally plausible while becoming semantically wrong. Table 4 reports the scene-complexity breakdown, including no-scene-change, single-scene-change, and multi-scene-change subsets, with both region (J) and contour (F) components shown alongside their mean. On this benchmark, the dual-agent framework improves $\mathcal{J}\&\mathcal{F}$ from 74.2 to 76.9, and the gain grows with scene complexity, reaching its largest value on multi-scene segments. This is exactly the signature we would expect from a failure-aware controller: the benefit is strongest where identity must be re-established across strong context shifts rather than merely refined.

Table 3 provides the mechanism-level view behind this benchmark gain. Table 3(b) summarizes controller activity: how often the verifier fires, how often the corrector is invoked, how many corrections are ultimately committed, and how these events break down by trigger type. Its 'Outcome' column reports directly verifiable committed corrections as better / worse / neutral relative to SAM3. The controller is sparse: over the full SecVOS run, it issues 1360 verifier calls, 240 corrector calls, and only 88 committed corrections, so the verifier is invoked on just 4.82% of unique video frames. Because SecVOS is annotated sparsely, not every verifier decision or commit is directly measurable at the intervention frame. On the 502 verifier decisions with a ground-truth box, the verifier attains 68.7% accuracy, 76.8% precision, and 60.0% recall, indicating a precision-oriented operating point suitable for guarded intervention. Among the 52 directly verifiable commits, 40 improve over SAM3, 11 worsen, and 1 is neutral, yielding 76.9% verified correction precision.

Table 3(a) complements this view by attributing committed interventions to the state of SAM3 at the moment of correction. The dominant positive regime is `sam3_missed`, where SAM3 has effectively lost the object and the commit restores a high-IoU state. The dominant negative regime is `sam3_was_ok`, where an unnecessary intervention overrides a correct track. Taken together, the two parts of Table 3 show that the improvement comes from selectively rescuing genuine failures rather than frequently rewriting the tracker state.

Table 4: Performance comparison on SecVOS-160 for different scene-complexity subsets.

Method	No Scene Change			Single Scene Change			Multi Scene Change			Overall
	J&F	*J*	*F*	*J&F*	*J*	*F*	*J&F*	*J*	*F*	*J&F*
XMem	71.9	72.0	71.8	47.0	47.9	46.2	41.9	42.4	41.4	48.4
DEVA	71.6	71.6	71.5	48.5	48.4	48.6	46.4	46.0	46.8	49.7
Cutie-base	72.5	72.2	72.8	53.0	52.9	53.2	48.3	47.8	48.9	52.7
SAM2.1	79.4	79.1	79.7	58.5	58.2	58.8	52.4	52.1	52.6	58.2
SAMURAI	81.8	81.6	81.9	60.6	60.6	60.7	59.3	58.9	59.7	62.2
SAM2.1Long	81.3	81.0	81.6	61.8	61.6	62.0	58.5	58.1	58.9	62.3
SeC	84.2	83.8	84.5	69.6	69.5	69.7	67.5	67.0	68.0	70.0
SAM3 baseline	87.09	87.0	87.2	74.62	74.7	74.6	69.27	69.1	69.4	74.21
Dual-agent (ours)	**88.11**	**87.9**	**88.3**	**76.22**	**76.3**	**76.1**	**73.39**	**73.3**	**73.5**	**76.94**

Table 5: SecVOS under dual-agent recipe: (a) trigger design, (b) voting, (c) refinement rounds.

(a) Trigger Ablation

Trigger setting	J&F	J	F
Occlusion	75.4	75.3	75.5
+ reappearance	75.6	75.5	75.7
+ effective-IoU variance	**76.9**	**76.9**	**77.0**

(b) Voting Ablation

Vote window	Vote stride	J&F	J	F
3	4	**76.9**	**76.9**	**77.0**
1	4	76.8	76.9	76.7
3	2	76.4	76.4	76.4

(c) Refinement Rounds

Rounds	J&F	J	F
3	**76.9**	**76.9**	**77.0**
1	75.6	75.6	75.7

Table 5 identifies which parts of the controller are responsible for the SecVOS gain. Table 5(a) is a trigger ablation: it starts from occlusion alone, then adds reappearance, and finally adds effective-IoU variance. The largest jump comes from effective-IoU variance, confirming that explicit drift sensing is the main driver of recovery volume. Table 5(c) is a refinement-depth ablation and shows that reducing correction from three rounds to one causes a comparable drop, so detecting failure is not enough unless the corrector is allowed to resolve it fully. By contrast, Table 5(b) shows that the voting design is comparatively stable: reducing the vote window from three to one changes little, while shrinking the vote stride is mildly harmful. On SecVOS, the load-bearing components are therefore triggered by design and sufficient correction depth, while voting mainly stabilizes the decision process. Overall, the framework improves SecVOS by +2.7 J&F while intervening on only 88 frames, showing that sparse, targeted recovery is sufficient to substantially improve long-horizon tracking.

Verification and commitment. Table 6(a) disentangles the roles of verification and commitment. Without verification, performance on SecVOS drops from 76.94 to 74.10 $\mathcal{J}\&\mathcal{F}$, essentially returning to the SAM3 baseline (74.21), despite retaining the commit gate. In contrast, removing the commit gate retains most of the improvement, reaching 75.90 $\mathcal{J}\&\mathcal{F}$ on SecVOS and 89.3 on LVOS-v2. This indicates that the gains cannot be explained by conservative acceptance alone: identifying when the tracker has failed and recovery is needed is a key component of the framework. We further analyze correction uncertainty and harmful interventions in Appendix H.

Computational cost. Table 6(b) quantifies the additional cost of failure-aware reasoning on a single H100. The full system requires 71.8 seconds per video, compared with 12.2 seconds for SAM3. FP8 verification reduces this to 56.2 seconds while maintaining nearly identical performance (76.8 vs. 76.9 $\mathcal{J}\&\mathcal{F}$). Reducing the occlusion polling frequency to once every 24 frames further lowers runtime to 47.7 seconds while retaining 76.41 $\mathcal{J}\&\mathcal{F}$. Thus, verification frequency and precision provide a direct trade-off between computational cost and tracking accuracy. The current implementation, however, is not real-time; additional analysis is provided in Appendix I.

Transfer beyond SAM3. We next examine whether the proposed framework depends on the SAM3 propagator. Without retraining, Dual-Agent improves SecVOS performance from 52.3 to 59.2 $\mathcal{J}\&\mathcal{F}$ with Cutie, from 58.4 to 65.1 with SAM2.1, and from 69.9 to 72.0 with SeC, corresponding to gains of 6.9, 6.7, and 2.1 points, respectively. These results show that the verification and recovery framework transfers across propagators with substantially different baseline performance, rather than being specific to SAM3. Further results are provided in Appendix F.

Single-object tracking. We further evaluate Dual-Agent on three long-term single-object tracking benchmarks. On TNL2K, Dual-Agent improves AUC from 65.2 to 66.5, with corresponding improve-

ments in precision (72.8 to 74.1) and normalized precision (74.3 to 75.7). On VOT-LT2022, recall increases from 0.574 to 0.613, while precision decreases from 0.632 to 0.610, yielding an overall F-score improvement from 0.601 to 0.612. On LaSOT-ext, the size-selected composite improves AUC from 63.7 to 64.3. Together, these results extend the evaluation beyond video object segmentation and show that the proposed failure-aware framework also benefits long-term object tracking. Details of the tracking protocol are provided in Appendix L.

Table 6: Verification and efficiency. (a) Verification and commit-gate ablations ($\mathcal{J}\&\mathcal{F}$); – denotes an unreported result. (b) SecVOS-160 cost on one H100, averaged per video; FP8 applies to the verifier.

(a) Verification and commitment

Configuration	SecVOS	LVOS-v2
SAM3	74.21	88.2
No verifier	74.10	–
No commit gate	75.90	89.3
Dual agent (full)	**76.94**	**89.4**

(b) Computational cost

Configuration	Seconds	PFLOP	$\mathcal{J}\&\mathcal{F}$
SAM3	12.2	1.378	74.21
Ours, bf16	71.8	2.22	**76.9**
Ours, FP8	56.2	2.22	76.8
FP8, poll every 24	47.7	2.01	76.41

Single-object tracking. On TNL2K, recovery improves AUC by 1.30 points (Table 7). On VOT-LT2022, higher recall offsets lower precision, improving F-score from 0.601 to 0.612. LaSOT-ext reports a size-selected composite rather than an unrestricted recovery policy (Appendix L).

Table 7: Box-initialized single-object tracking. LaSOT-ext and TNL2K scores are percentages; VOT-LT2022 uses precision, recall and F-score. †Size-selected composite; protocols and initialization comparisons are specified in Appendix L.

LaSOT-ext				**TNL2K**				**VOT-LT2022**			
Method	AUC	P	P_{norm}	Method	AUC	P	P_{norm}	Method	Pr	Re	F
SAM 3	63.7	74.8	72.9	SAM 3	65.2	72.8	74.3	SAM 3	**0.632**	0.574	0.601
Dual agent	**64.3**†	**75.5**†	**73.5**†	Dual agent	**66.5**	**74.1**	**75.7**	Dual agent	0.610	**0.613**	**0.612**

Qualitative Results. Figure 3 illustrates two representative failure modes recovered by the proposed dual-agent framework. The top sequence shows long-horizon *reappearance*: after the target leaves the scene and re-enters with significant pose and motion change, SAM3 fails to recover it. The verifier detects the propagated prediction as inconsistent with the target, after which the corrector re-localizes the object and restores stable propagation. The bottom sequence shows *drift in cluttered scenes*. As the target train moves through visually similar distractors, SAM3 latches onto a neighboring object and its propagated state becomes unstable. The effective-IoU variance trigger activates the verifier, which confirms the identity failure. The corrector then recovers the correct target and rewrites the tracker state, allowing propagation to continue successfully.

5 Conclusion

We present a dual-agent framework that uses SAM3 as the propagator and augments it with explicit verification and selective recovery. By separating reliability estimation from target correction, the framework turns a SAM3-style propagator into a failure-aware tracking system. The verifier reasons over short spatio-temporal context to detect unreliable predictions, while a learned correction agent recovers the target only when intervention is warranted. Evaluation across segmentation and tracking benchmarks demonstrates that robust tracking requires not only stronger propagation, but also explicit reasoning about when propagation should no longer be trusted. One limitation of our framework is its reliance on the reliability of the verification agent. In highly ambiguous scenarios, incorrect reasoning may lead to missed failures or unnecessary corrections, while the additional verification stage also increases computational overhead. The implementation is not real-time; transfer to specialized domains and video-level disjointness from VLM pretraining remain unverified (Appendix M).

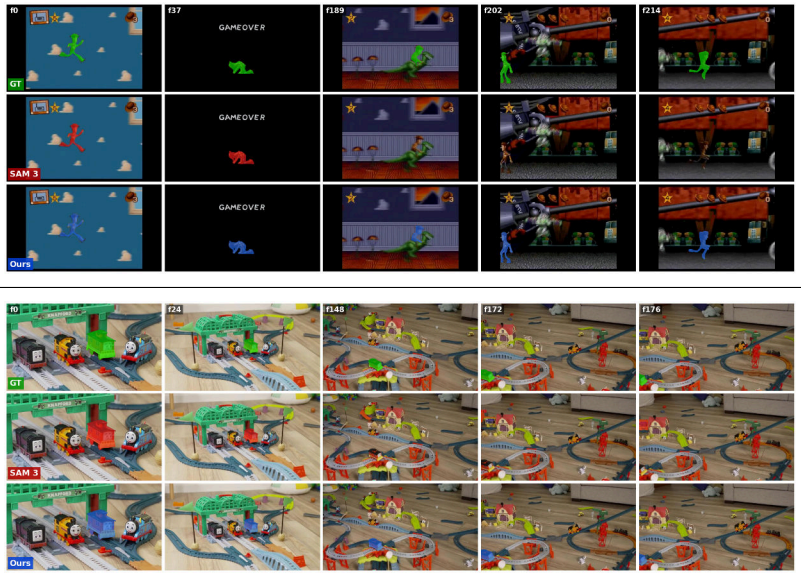


Figure 3: Qualitative recovery on two SAM3 failure modes. **Top:** reappearance after a long absence; **Bottom:** drift on the target toy train in a cluttered scene with similar distractors. For each example, rows show ground truth (top), the SAM3 baseline (middle), and our dual-agent framework (bottom).

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A Implementation Details

Verification agent. We use Qwen3-VL-8B-Thinking in its native thinking mode with no fine-tuning, prompted with a structured tracking template. Each verifier query is a 7-image visual context laid out as 2 ground-truth exemplar frames, 2 uniformly sampled context frames, 2 recent context frames at stride 4, and 1 query frame. The verifier votes 3 times per call with a vote stride of 4 frames; the majority verdict is committed. In our experiments, we set $\tau_{\text{conf}} = 0.7$, $K = 3$, $M = 12$ and $N = 8$.

Correction agent. We use Qwen3-VL-4B-Instruct adapted with LoRA (rank 16, $\alpha = 32$, dropout 0.05, applied to all linear layers; vision tower frozen, and keep the multimodal projector trainable). The corrector is fine-tuned with AdamW at learning rate 2×10^{-5} on a cosine schedule with warmup ratio 0.05 and weight decay 0.01, for 2 epochs at effective batch size 16 (4 GPUs $\times$ batch 2 $\times$ gradient-accumulation 2), with bf16 mixed-precision training, gradient checkpointing, a 16,384-token sequence cutoff, and a 401,408-pixel image budget matched to inference. We fine-tune the VLM on 4 H100s for around a day. The corrector takes the same 7-image layout as the verifier and is invoked in no-think mode using a single-key visible-plus-bounding-box prompt template.

Selective triggering. The closed-loop controller fires the verifier on three signals applied in disjunction: occlusion polling (one query every 12 consecutive occluded frames), reappearance probing (one query when the propagator recovers a target after an absence, with cooldown 8 frames), and a streaming effective-IoU variance trigger (with a 4-frame window, threshold 0.15, persistence 2, and a minimum spacing of 8 frames between consecutive fires). When the verifier returns "incorrect," the corrector is invoked with up to 3 retry rounds and its output is gated by an effective-IoU guard at 0.7 against the propagator's current state.

Propagator. SAM3 remains entirely frozen across all experiments; the dual-agent framework operates as an external control loop that selectively rewrites SAM3's state when the verifier flags a failure.

B Dataset Details

This appendix gives the full inventory of every benchmark and development subset referenced in the main paper, including sequence and object counts, annotation density, evaluation protocol, and the exact subset definitions used for ablations. All numbers correspond to the splits we actually ran and evaluated; sequence-list files used for the locked subsets are provided alongside the supplementary as plain-text manifests.

B.1 Benchmark Inventory

Table 8: Benchmarks and splits used in the paper. The annotation column distinguishes segmentation masks from tracking boxes; all tracking benchmarks use first-frame box initialization.

Benchmark	Split	Sequences	Objects	Frames / annotations	Use
LVOS-v2	valid	140	239	every frame	full benchmark
SecVOS	full	160	182	sparse multi-frame masks	full benchmark
LaSOT-ext	extension	150	150	per-frame boxes	tracking
TNL2K	test	700	700	per-frame boxes	tracking
VOT-LT2022	long-term	50	50	boxes / absence	tracking

LVOS-v2. We evaluate on the official LVOS-v2 validation split, which consists of 140 long sequences containing 239 annotated objects in total. Annotations are dense: every video frame has a corresponding ground-truth mask. The benchmark is designed to stress long-horizon memory: sequences are substantially longer than those in DAVIS or YouTube-VOS, so propagation errors compound over many frames. We report results on the standard LVOS-v2 evaluation harness (`lvos-evaluation`), which produces $\mathcal{J}_s$, $\mathcal{F}_s$, $\mathcal{J}_u$, $\mathcal{F}_u$, and overall $\mathcal{J}\&\mathcal{F}$ on the seen / unseen splits.

SecVOS. SecVOS is a 160-sequence benchmark with 182 annotated objects, designed around multi-shot discontinuity, scene transitions, and target reappearance across semantically distinct contexts. We use the standard SecVOS evaluation protocol from the SeC paper, which scores J&F

on the full benchmark and additionally stratifies the score into three scene-cut buckets: *no scene change*, *single scene change*, and *multi scene change*. We run this stratification using the SeC-provided evaluator (`sav_evaluator.py` with `-scene_info`) so that results are directly comparable to numbers reported in the SeC paper. Annotations on SecVOS are sparse in time, similar in spirit to SA-V: a non-trivial fraction of dual-agent corrections in our framework fire between annotated frames, and the direct attribution analysis in the main paper uses only commits with a ground-truth box at the intervention frame. The 36 commits without such a box are excluded from the reported localization-precision denominator.

Single-object tracking benchmarks. LaSOT-ext evaluates long-term tracking on 150 extension sequences, TNL2K on its 700-sequence test split, and VOT-LT2022 on 50 long-term sequences with target disappearance and reappearance. Each sequence contains one target, specified by its first-frame bounding box; TNL2K language descriptions are not used. LaSOT-ext and TNL2K use success AUC, 20-pixel precision, and normalized precision at 0.2. VOT-LT2022 uses the official long-term precision, recall, and F-score protocol without re-initialization. Appendix L details the sequence-size selection for LaSOT-ext, VOT confidence guard, and initialization-specific comparisons.

C Training Data Details

The correction agent is trained on a single corrector-SFT corpus that we generate ourselves from SAM3 propagation rollouts on SA-V training videos.

C.1 Dataset Identity and Sizes

The corpus is split sequence-stratified at generation time into a 95% train shard and a 5% validation shard, yielding 75,341 training samples and 4,116 validation samples for a total of 79,457 supervised correction examples. The split is deterministic (`seed=42`) and stratified by source video so that no SA-V sequence appears in both shards: this prevents the corrector from receiving an inflated validation score by memorizing per-video patterns. Because the corpus is sequence-stratified rather than frame-stratified, validation-set samples test temporal generalization within unseen sequences from the same SA-V shards.

C.2 Per-Sample Schema

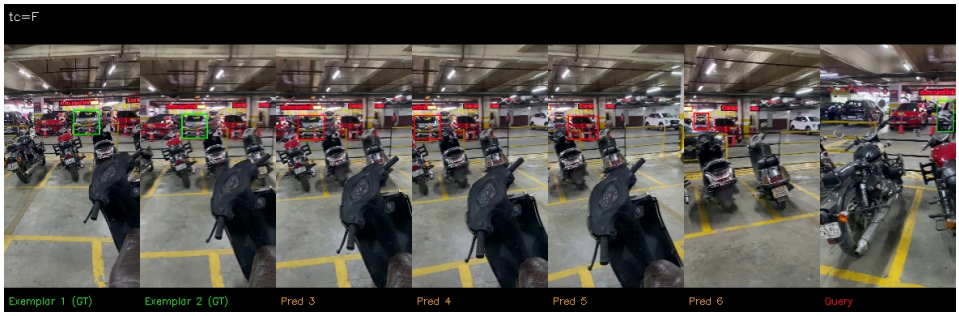


Figure 4: An example from the dataset.

Each supervised example encodes a closed-loop correction decision. As shown in Figure 4, the visual context is laid out as a fixed seven-frame window:

- **2 ground-truth exemplar frames** drawn from the early part of the sequence, where the target identity is unambiguous.
- **2 uniform context frames** sampled at uniform stride across the elapsed sequence to capture long-range appearance changes.
- **2 recent context frames** immediately preceding the query, capturing short-term motion and occlusion state.

- **1 query frame** at which the corrector must predict whether the target is visible and, if so, where.

Each frame is annotated with an enrichment payload: SAM3's predicted box at that frame (or an explicit "no detection" marker), the ground-truth box at that frame (when an annotation exists), and the IoU between the two. The supervised target at the query frame is a structured JSON object containing a `visible` boolean and, when visible, a normalized `box_2d`. Outputs are produced in a no-think format with the `<think>` block stripped at generation time.

C.3 Generation Pipeline

Samples are synthesized by a single deterministic generator script operating over the cached SAM3 tracking results. The generator's behavior is determined by parameters listed in Table 9; these flags are the entirety of the locked configuration that produced the corpus reported in this paper. No flags marked here as locked were varied during dataset generation.

Table 9: Locked sample-generation arguments for the corrector-SFT corpus. All probabilities are per-sample independent draws.

Argument	Value
`correct_ratio`	0.1
`id_switch_context_prob`	1.0
`idsw_query_switch_prob`	0.3
`max_samples_per_object`	10
`max_not_visible_ratio`	0.4
`seed`	42

These knobs serve specific roles. `correct_ratio=0.1` controls the fraction of samples on which SAM3 is already correct and the supervised label simply mirrors the ground-truth box rather than overriding it; this small but nonzero share is what prevents the corrector from collapsing into an unconditional "override SAM3" policy at deployment. `id_switch_context_prob=1.0` turns identity-switch context scaffolding on for every sample, so that on every example the corrector additionally sees a variant in which one or two recent red boxes are replaced by a mined distractor box from the same frame, forcing it to anchor identity on the green exemplars rather than on the most recent red context. `idsw_query_switch_prob=0.3` additionally swaps the query-frame red box for a distractor on 30% of identity-switch samples, simulating a tracker that has followed the distractor through both context and query. `max_samples_per_object=10` caps the number of samples drawn from any single (sequence, object) pair, preventing long, hard-to-track objects from dominating the corpus. `max_not_visible_ratio=0.4` caps the fraction of fully-occluded query frames so that the corrector does not collapse into an "always not visible" degenerate solution. `seed=42` fixes the dataset partition and all stochastic choices. Identity-switch context is the only stochastic augmentation in this corpus.

C.4 Dataset Analysis

We characterize the corrector-SFT corpus along seven axes that together describe what the corrector is supervised to learn: (i) the distribution of failure modes the corrector must handle, (ii) the joint distribution of SAM3 correctness and target visibility, (iii) the quality of the visual context the corrector reasons over, (iv) the augmentation budget, (v) the per-sequence and temporal coverage, (vi) the geometric scale of supervised targets, and (vii) the input-quality distribution at the query frame. All numbers are produced by the project's dataset-distribution analyzer running on the exact deployed corpus. Because the deployed sharegpt-format files used by the trainer drop the per-sample provenance fields (each sample retains only `messages`, `images`, and a small enrichment metadata block), we run the analyzer on the equivalent reinforcement-learning-format export, whose 71,597-sample train shard plus 7,860-sample validation shard sum to the same 79,457 examples and preserve the `failure_type`, `is_sam3_correct`, and context-frame metadata produced at generation time. The two formats are bit-identical at the example level; only the train/val partition differs.

C.4.1 Failure-Mode Composition

The corrector is trained to recognize and recover from a small set of structurally distinct failure modes that the propagator exhibits at deployment. Table 10 reports the partition of the corpus by failure type; the eleven categories are mutually exclusive and together account for the full 79,457 samples.

The composition is engineered, not incidental. Roughly two-thirds of the corpus targets the override regime: `false_negative` (17.8%) supervises the corrector to localize from context when SAM3 has produced no box at all, `low_iou` (15.1%) supervises refinement when SAM3 has drifted, `perturbed` (15.5%) trains the corrector to remain stable when its own context input is jittered, and three identity-switch variants (8.0% combined) train it to resist confusion from co-occurring distractor objects. The remaining samples balance this with non-override supervision: `correct` (14.6%) teaches the corrector to mirror SAM3 when SAM3 is already right, `both_none` (13.5%) teaches graceful behavior when there is nothing to track, and the two negative-label categories (`false_positive` 4.4% and `synthetic_fp` 6.0%) train suppression of spurious detections. The deliberate inclusion of mirror-SAM3 supervision is what prevents the corrector from collapsing into an "always override" degenerate solution at deployment time, where it would pollute the propagator's memory with redundant rewrites of already-correct frames.

Table 10: Sample-type distribution of the corrector-SFT corpus (79,457 samples), produced by the project's distribution analyzer.

Sample type	Count	Share	Supervised behavior
`false_negative`	14,131	17.8%	emit a fresh visible-true box from context
`perturbed`	12,349	15.5%	recover from jittered context boxes
`low_iou`	12,011	15.1%	emit a refined box over a drifted SAM3 box
`correct`	11,618	14.6%	mirror SAM3's already-correct prediction
`both_none`	10,746	13.5%	emit visible-false on absent target
`synthetic_fp`	4,807	6.0%	reject artificially-injected false positive
`paired_occluded`	4,000	5.0%	decision on occlusion-onset pairs
`false_positive`	3,478	4.4%	emit visible-false to suppress SAM3 hallucination
`id_switch (false_negative)`	2,926	3.7%	resist distractor when SAM3 missed the target
`id_switch (low_iou)`	2,753	3.5%	resist distractor when SAM3 drifted
`id_switch (false_positive)`	638	0.8%	resist distractor when SAM3 hallucinated
Total	79,457	100%	

C.4.2 Joint Distribution of SAM3 Correctness and Visibility

A coarser but more diagnostic view of the corpus is the cross-tabulation of the supervised label (whether SAM3 was correct at the query frame) against the query-frame visibility (whether the target is actually there). Table 11 reports this two-by-two split.

Two observations follow from this table. First, the marginal class balance — 33.5% correct versus 66.5% incorrect — indicates that the corrector sees twice as many failure cases as it sees easy mirror cases, which is the right proportion for a corrector whose job is to repair propagation rather than describe it. Second, the cross-tabulation reveals a subtle but deliberate skew: when the target is visible, SAM3 is correct only 22.2% of the time, but when the target is absent, SAM3 is correct 60.1% of the time. In other words, the corpus oversamples the regime where SAM3 has produced a confident-looking box on a frame whose target is missing or substantially wrong, which is exactly the regime where override has the largest expected payoff. The smaller correct-on-absent slice (17.9% of the corpus) is what teaches the corrector to recognize correct-by-default behavior and avoid spurious overrides on absent targets.

C.4.3 Quality of the Visual Context

The corrector reasons over a fixed seven-frame visual context (two early-sequence ground-truth exemplars, two uniformly sampled context frames, two recent context frames, and one query frame), and the quality of this context directly determines what reasoning the corrector can perform. We characterize context quality at three granularities: per frame, per sample, and per failure type.

Table 11: Class balance (SAM3 correct / incorrect) cross-tabulated against query-frame visibility, on all 79,457 samples. Rows are the supervised label; columns are the visibility of the target at the query frame.

SAM3 was …	Visible target	Absent target	Total
Correct	12,385 (15.6%)	14,223 (17.9%)	26,608 (33.5%)
Incorrect	43,403 (54.6%)	9,446 (11.9%)	52,849 (66.5%)
Total	55,788 (70.2%)	23,669 (29.8%)	79,457 (100%)

Table 12 reports the per-frame IoU histogram across all 249,905 non-exemplar context frames in the corpus. The distribution is strongly bimodal: 43.0% of context frames are in the high-quality bucket $[0.95,\ 1.01)$ where SAM3 has accurately tracked the target, and 18.2% are in the lowest bucket $[0.0,\ 0.10)$ where SAM3 has effectively lost it. Only 23.4% of frames fall into the wide middle band $[0.30,\ 0.85)$, indicating that context quality is rarely ambiguous — frames are either clearly clean or clearly degraded. This bimodality is what enables the corrector to learn from a relatively small training corpus: most ambiguity is in the small middle band rather than spread across the IoU axis.

Aggregating per-frame IoU into a per-sample difficulty label (Table 13) gives a complementary view of how challenging each training example is. Almost two-thirds of samples (64.1%) are *easy* in the sense that the corrector has at least two clean context frames to anchor on; one in six (16.8%) is *hard* with only a single good context frame or a single ground-truth-corrected frame; 3.8% are *noisy* with no good context and no ground-truth correction; and 15.3% have *no usable context at all*, forcing the corrector to rely entirely on the two exemplar frames. The non-trivial *no-context* fraction is intentional: at deployment, the corrector is sometimes called immediately after a long occlusion when the recent context is mostly empty, so the corpus must contain examples where the exemplars are the only source of identity information.

Stratifying mean context IoU by failure type (Table 14) reveals where context is hardest. The mean drops monotonically from 0.924 on `correct` samples (where SAM3 is right and context is clean) to 0.272 on `id_switch` samples, where the seven-frame window is constructed to include a confounding object. Because identity-switch supervision is delivered with deliberately noisy context, the corrector cannot rely on a single "trust the recent box" heuristic for these cases — it must use the early-sequence exemplars to anchor identity. The `low_iou` (0.536) and `false_positive` (0.495) categories sit in the middle, supervising the corrector to override SAM3 even when context tokens themselves provide a noisy version of the failed prediction.

Table 12: Histogram of context-frame SAM3 IoU against ground truth across all 249,905 non-exemplar context frames in the corpus. Mean IoU 0.683, median 0.925, standard deviation 0.388.

IoU bucket	Frames	Share
$[0.00,\ 0.10)$	45,598	18.2%
$[0.10,\ 0.30)$	15,763	6.3%
$[0.30,\ 0.50)$	13,595	5.4%
$[0.50,\ 0.70)$	11,646	4.7%
$[0.70,\ 0.85)$	15,922	6.4%
$[0.85,\ 0.95)$	39,947	16.0%
$[0.95,\ 1.01)$	107,434	43.0%

Table 13: Per-sample difficulty distribution, classified by how many usable context frames the sample contains.

Difficulty	Samples	Share	Definition
easy	50,948	64.1%	at least 2 context frames with IoU > 0.5
hard	13,357	16.8%	one good context frame or one GT-corrected frame
noisy	3,000	3.8%	no good context frame, no GT correction
no_context	12,152	15.3%	all context boxes null or absent

Table 14: Mean context-frame SAM3-vs-ground-truth IoU stratified by sample type (excluding the exemplar frames). Lower mean IoU corresponds to noisier context the corrector must reason over.

Sample type	N	Mean IoU	Median IoU
`correct`	9,531	0.924	0.966
`both_none`	9,723	0.784	0.888
`perturbed`	9,736	0.774	0.895
`synthetic_fp`	4,257	0.762	0.873
`false_negative`	9,989	0.749	0.844
`paired_occluded`	3,365	0.724	0.811
`low_iou`	10,996	0.536	0.511
`false_positive`	3,391	0.495	0.458
`id_switch`	6,317	0.272	0.316

C.4.4 Sequence and Temporal Coverage

The corpus is drawn from 5,145 unique source sequences and 16,045 unique annotated objects across SA-V training shards `sav_000` through `sav_005`. Table 15 reports the per-sequence and per-object sample-count statistics; both distributions are strongly right-skewed. The median sequence contributes 9 samples and the median object contributes 2, but the most heavily sampled sequence contributes 155 samples and the most heavily sampled object contributes 36. A per-object cap (`max_samples_per_object`=10) bounds the per-object skew below what a purely failure-driven sampling would produce, which is what prevents long, hard-to-track objects from dominating the corpus and biasing the corrector toward their idiosyncrasies.

Temporal distances from the query frame (Table 16) characterize how far back the context window reaches. Recent context frames are concentrated close to the query (mean 72 frames, median 24), while exemplar frames are sampled from the early part of the sequence and are therefore much further away from the query (mean 143 frames, median 128) with maxima close to 1,000 frames. The asymmetry between recent context and exemplars matches the role each plays in the prompt: recent context provides short-term motion and occlusion cues, while exemplars provide a stable identity anchor that does not depend on whatever the propagator has been doing in the meantime.

Table 15: Per-sequence and per-object coverage of the 79,457-sample corpus, drawn from 5,145 unique sequences and 16,045 unique annotated objects across SA-V training shards `sav_000`–`sav_005`.

	Mean	Median	Min	Max
Samples per sequence	15	9	1	155
Samples per object	5	2	1	36

Table 16: Temporal-distance statistics (in source-video frames) from the query frame, across all 79,457 samples.

Distance from query frame	Mean	Median	Max
Context-to-query	72	24	940
Exemplar-to-query	143	128	948

C.4.5 Geometric Scale of Supervised Targets

Among the 55,788 samples with a visible target, the ground-truth box occupies on average 9.0% of the frame area and 3.5% at the median. Table 17 reports the full size-bucket distribution.

The distribution skews sharply toward small targets: 59.5% of supervised boxes are below 5% of the frame area and 82.9% are below 15%. Only 7.4% of boxes occupy more than 30% of the frame. This skew is a direct consequence of using SA-V training shards as the source: SA-V is dominated by everyday-object videos in which the target rarely fills the frame. The skew is also why we caution against transferring a corrector trained on SA-V to applications where targets routinely occupy a large

fraction of the frame — the corrector has seen comparatively few training examples in the upper size buckets and may be miscalibrated for them. For the benchmarks reported in the main paper (SA-V sav-test, LVOS-v2, MOSE, SecVOS), all of which contain similarly skewed size distributions, the corpus is well-matched.

Table 17: Ground-truth box size distribution across the 55,788 visible-target samples, expressed as a fraction of frame area.

Box area as % of frame	Samples	Share
$[0\%,\ 1\%)$	8,630	15.5%
$[1\%,\ 5\%)$	24,574	44.0%
$[5\%,\ 15\%)$	13,079	23.4%
$[15\%,\ 30\%)$	5,380	9.6%
$[30\%,\ 60\%)$	3,119	5.6%
$[60\%,\ 100\%]$	1,006	1.8%

C.4.6 Image-Count Distribution

The canonical per-sample layout uses seven images (two ground-truth exemplars, two uniform context frames, two recent context frames, and one query frame). Not every sample can fill this layout: short sequences in SA-V do not always have enough elapsed frames to admit two recent and two uniform context frames distinct from the exemplars. Table 18 reports the distribution of image counts per sample.

The vast majority of samples (72.9%) use the canonical seven-image layout. The next most common configuration is three images (13.2%), corresponding to short sequences in which only one exemplar plus one context frame plus the query are available. Mixed shorter configurations (four to six images) account for 11.7% combined. Two-image samples (2.2%) are the degenerate case in which only the exemplar and the query exist. We deliberately do not pad short samples to seven images: a placeholder image would either repeat content the model has already seen or introduce a synthetic out-of-distribution token. Allowing variable image counts at training time prevents the corrector from overfitting to the seven-image template and makes inference robust to short sequences in deployment.

Table 18: Image-count distribution per sample. The canonical seven-image layout is 2 exemplars + 2 uniform context + 2 recent context + 1 query.

Images per sample	Samples	Share
2	1,713	2.2%
3	10,477	13.2%
4	2,772	3.5%
5	4,376	5.5%
6	2,169	2.7%
7 (canonical)	57,950	72.9%

C.4.7 Input Quality at the Query Frame

The most direct measure of the input distribution the corrector must learn from is the IoU between SAM3's prediction and the ground truth at the query frame, computed across the entire corpus. Table 19 reports the full ten-bin histogram for both `mask_iou` and `bbox_iou`; the two metrics agree closely, so we discuss them together.

Aggregate statistics are stark: mean IoU is 0.208, median is 0.000, and standard deviation is 0.330. The histogram is strongly bimodal, with 59.8% of samples in $[0.0,\ 0.1)$ and 12.2% in $[0.9,\ 1.0)$, while every intermediate bin holds less than 10%. Two design implications follow. First, the corrector's expected per-frame value-add is very large at training time: any reasonable corrector that lifts the bottom-bin samples even modestly produces a substantial absolute IoU gain, which is what enables the +30 percentage-point lift reported in the Results Section of the main paper. Second, the bimodality implies the corrector should naturally specialize into two regimes — a high-confidence mirror behavior on the right mode and an aggressive override on the left mode — without a wide

intermediate band where calibration is ambiguous. This matches what we observe in the results: the corrector's accepted commits cluster at large per-frame IoU lifts rather than at marginal refinements.

Table 19: Histogram of SAM3-vs-ground-truth bounding-box IoU at the query frame across all 79,457 samples. Both `mask_iou` and `bbox_iou` are reported; the two agree very closely. Absent-target and false-negative samples contribute IoU 0 by construction.

IoU bucket	mask_iou samples	Share	bbox_iou samples	Share
$[0.0,\ 0.1)$	47,455	59.7%	47,490	59.8%
$[0.1,\ 0.2)$	7,026	8.8%	7,702	9.7%
$[0.2,\ 0.3)$	4,904	6.2%	5,392	6.8%
$[0.3,\ 0.4)$	3,956	5.0%	3,874	4.9%
$[0.4,\ 0.5)$	3,759	4.7%	2,614	3.3%
$[0.5,\ 0.6)$	762	1.0%	752	0.9%
$[0.6,\ 0.7)$	506	0.6%	555	0.7%
$[0.7,\ 0.8)$	360	0.5%	380	0.5%
$[0.8,\ 0.9)$	1,301	1.6%	1,040	1.3%
$[0.9,\ 1.0)$	9,428	11.9%	9,658	12.2%

D Verification Agent Prompt

Our verification agent is a Qwen3-VL-8B-Thinking model, which receives the following prompt:

```
You are assessing a single-object tracker. You see VIDEO FRAMES with overlays:
- First exemplar frame(s): GREEN box marks the TARGET object's true appearance.
- Context frames: RED box = tracker prediction (may be wrong); no RED box = tracker says target absent.
- Query frame (last): RED box (or none) = the prediction you must assess.

Describe the REAL OBJECTS inside the boxes, never the box colour itself. Reason from what you SEE in
     the boxes.

NOTES ON THE FRAME LIST:
- (t=N) is the source video-frame index. The same (t=N) may legitimately appear twice if an exemplar
     coincides with a context sample - treat them as the SAME physical frame and move on.
- "earlier context" frames are far in the past, "recent context" frames are just before the query.

YOUR TASK on the QUERY frame is the discrimination question:
  "Does the tracker correctly track the GREEN target instance in this query frame?"

Answer it by following these steps. Do NOT skip step (a):

(a) LOCATE THE TARGET IN THE QUERY IMAGE FIRST.
    Before looking at the red box at all, scan the query image and find where the
    target object (the one in the GREEN exemplar) actually is. Possible outcomes:
      (a.1) Target is fully visible somewhere in the query.
      (a.2) Target is partially visible - some of it is out-of-frame, occluded by a
            person/object/the frame edge, or behind something. Note WHICH PART is
            visible (head only? legs only? a thin sliver at the edge?).
      (a.3) Target is genuinely absent - fully occluded, fully out-of-frame, gone.
            ONLY pick this if you cannot find the target anywhere in the image.

(b) NOW LOOK AT THE TRACKER OUTPUT.
    Two cases:
      Case I - A RED box IS shown on the query:
        * If (a.1) or (a.2): does the red box overlap the target's currently-visible
          extent by at least HALF? Tight head-only / legs-only / thin-sliver crops
          are CORRECT *if and only if* they sit on the visible part of the target.
          A red box on empty floor / sky / wall / a different object (e.g. a balloon,
          a shoe, another person, the ground) when the target is visible elsewhere
          → INCORRECT (this is an ID switch / drift / false positive even though
          the box exists). State explicitly what is inside the box.
        * If (a.3): the tracker drew a box when the target is absent → INCORRECT
          (false positive).
      Case II - NO RED box on the query (the prompt says "no tracker prediction
      shown"):
        * If (a.1) or (a.2): the target IS visible but the tracker abstained →
          INCORRECT (missed visible target).
        * If (a.3): correct abstention → CORRECT.

(c) VERDICT.
    Be DECISIVE. Default to CORRECT only when you genuinely cannot tell.
    Flag INCORRECT whenever step (b) produced a clear, named failure - even if it
    feels harsh. The downstream corrector handles the actual fix.

EDGE-CASE NOTES:
- Edge-of-frame thin RED boxes (touching x=0 or y=0 or a frame edge) usually mean
  the target is exiting the frame. If you saw the target moving toward that edge
  in the recent context, the tiny box at the edge is almost certainly STILL ON
  the partially-visible target → CORRECT. Do not dismiss it as "on empty space"
  unless you can confirm the target is not visible at that edge at all.
- A RED box that sits on a clearly different object (different colour, different
  category, distractor that crossed paths with the target, the ground) while
  the target is visible elsewhere in the frame → INCORRECT (ID switch / drift).
- Imperfect tightness on a fully-visible target (box covers ~70-90% of the
  target) → CORRECT. The corrector cares about IoU>=0.5, not pixel-perfect.

OUTPUT FORMAT (MANDATORY - your response is parsed automatically):
Your final line MUST be of the form:
<answer>{"tracking_correct": true}</answer>
or
<answer>{"tracking_correct": false}</answer>
Do NOT produce any other JSON keys. Do NOT continue reasoning after </answer>.

LENGTH BUDGET: keep your `<think>` block under ~250 words. If you find yourself
rephrasing the same observation, STOP and emit your verdict.

Decision rules summary:
- TRUE: red box covers ≥50% of the target's visible extent, OR no box AND target genuinely absent.
- FALSE: target visible but no box; box on a different object; box on empty space when target is
     visible elsewhere; box covering well under half of a fully-visible target.

Case 1 - Tracking is correct:
<think>your reasoning here</think>
<answer>{"tracking_correct": true}</answer>

Case 2 - Tracking is incorrect:
<think>your reasoning here</think>
<answer>{"tracking_correct": false}</answer>
```

E Further Qualitative Results

More visual tracking outputs are illustrated in Figure 5.

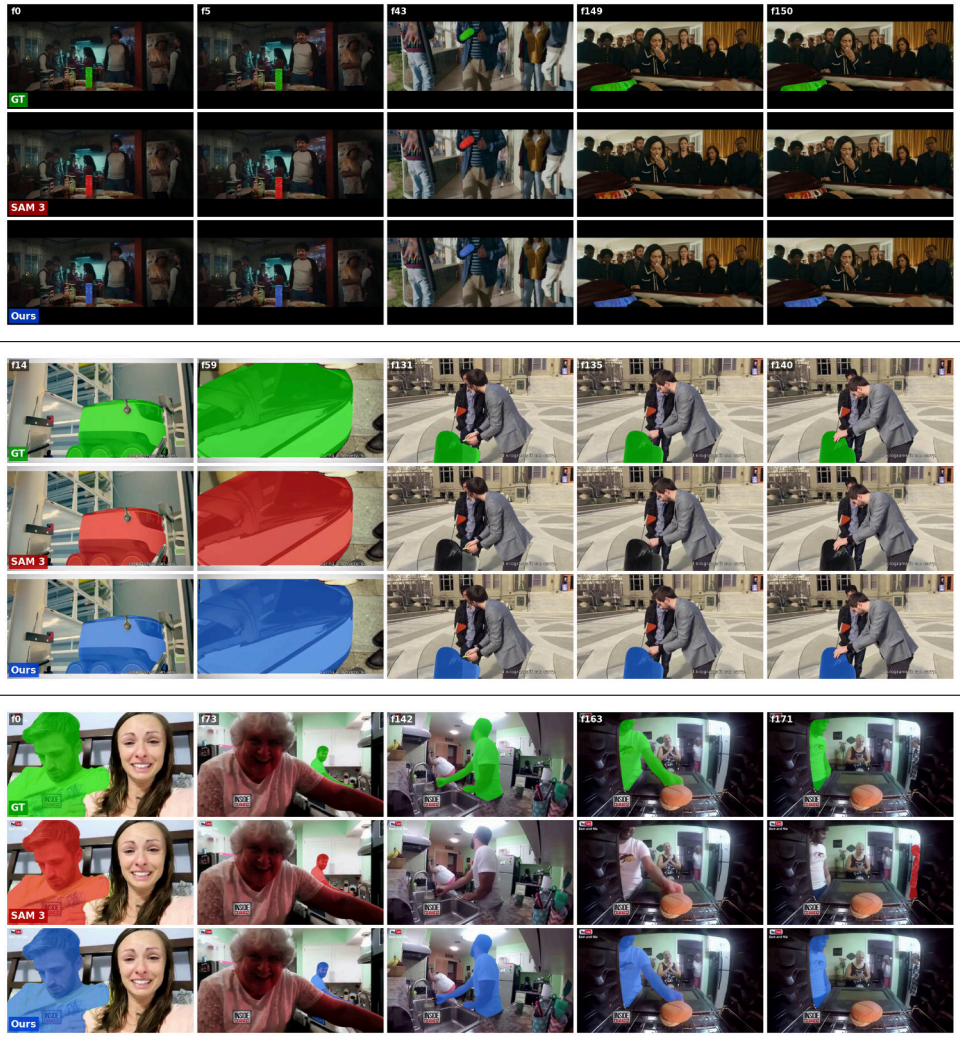


Figure 5: Further qualitative examples of our model vs. SAM3. **Top:** AS the target object is partially reappeared after a long time in a largely different context, SAM3's score for the object drops, and it fails to identify it; this triggers our recovery mechanism, and the object is re-identified by our dual agents. **Middle:** Upon partial appearance of the object, SAM3 fails to find the target object, due to its significant shift from its prior appearance. However, it is recovered by a further VLM-based retrieval through our designed agents. **Bottom:** Due to the change of the view of the scene, a different pose of the person appears in the video, resulting in the object not being found by SAM3. Our verification agent is thus triggered and detects the need for correction, with the recovery applied afterwards by the second correction agent. For each example, rows show ground truth (top), the SAM3 baseline (middle), and our dual-agent framework (bottom).

Table 20: Transfer of the unchanged dual-agent layer to other frozen propagators on SecVOS-160 ($\mathcal{J}\&\mathcal{F}$, official scene-stratified protocol). No component is retrained; only the propagator is replaced.

	Overall		No scene change		Single change		Multiple changes	
Propagator	alone	+dual agent	alone	+dual agent	alone	+dual agent	alone	+dual agent
Cutie [6]	52.3	**59.2** (+6.9)	72.5	**75.9** (+3.4)	53.2	**59.5** (+6.3)	47.4	**56.6** (+9.2)
SAM2.1 [19]	58.4	**65.1** (+6.7)	79.4	**83.4** (+4.0)	58.5	**61.6** (+3.1)	52.4	**61.3** (+8.9)
SeC [28]	69.9	**72.0** (+2.1)	84.1	**84.6** (+0.5)	69.7	**72.2** (+2.5)	67.3	**70.5** (+3.2)
SAM3 [4]	74.2	**76.9** (+2.7)	87.1	**88.1** (+1.0)	74.6	**76.2** (+1.6)	69.3	**73.4** (+4.1)

Table 21: Verification and commitment ($\mathcal{J}\&\mathcal{F}$). (a) Bypassing verification returns SecVOS-160 to the frozen-propagator level. (b) Removing the commit gate keeps most of the gain on both benchmarks.

(a) Agent ablation

Configuration	SecVOS-160
SAM3 alone	74.21
Corrector at every trigger (verifier skipped)	74.10
Dual agent (ours)	**76.94**

(b) Commit-gate ablation

Configuration	Gate	SecVOS-160	LVOS-v2
SAM3 alone	–	74.21	88.2
Dual agent	off	75.90	89.3
Dual agent (ours)	on	**76.94**	**89.4**

F Transfer Across Propagators

SAM3 provides residual failures of a strong propagator for corrector training. To test whether recovery is tied to these trajectories, we replace only the propagator and retain the zero-shot verifier and the trained corrector without adaptation. Table 20 reports the corresponding standalone and dual-agent runs on SecVOS-160. The transfer-run baselines are measured for these experiments and can differ slightly from the published comparison rows in the main paper. Cutie does not expose the confidence signal used by the other propagators, so only occlusion polling, every 12 frames, is used for this backbone.

The layer improves each evaluated propagator, with larger gains on the weaker Cutie and SAM2.1 baselines. Improvements on SeC indicate that external verification and recovery can complement semantic reasoning within a propagator. The largest gains occur after multiple scene changes.

G Verification and Commit-Gate Ablations

We separate failure verification from correction acceptance with two ablations. First, we bypass the verifier and invoke the corrector at each trigger, retaining the downstream commit checks. Second, we disable the commit gate while retaining verification and correction. Table 21 shows that removing verification eliminates the SecVOS gain, whereas disabling the gate retains most of the improvement on both benchmarks. This comparison supports a contribution from the verify-and-recover loop beyond conservative acceptance checks.

H Correction Reliability and Harmful Interventions

Precision and its uncertainty. The reported localization precision is conditional on a nonempty ground-truth box at the intervention frame. It is the fraction of these commits whose box IoU improves over the propagator, with tolerance 10^{-6}. Commits without a ground-truth box are excluded rather than counted as successes; this statistic does not measure the correctness of absence decisions. Table 22 reports Wilson binomial confidence intervals and object-clustered bootstrap intervals. The latter resample object tracks, preserving dependence among commits on an object. These intervals quantify finite-evaluation-sample uncertainty, not variability across independently trained correctors.

The SAM3 interval remains wide because only 52 of its 88 commits have a ground-truth box at the intervention frame. The same corrector yields high precision on the larger evaluable sets for SAM2.1

Table 22: Correction precision on SecVOS-160. Confidence intervals are 95%; all precision values are percentages.

Propagator	Commits	Evaluated	Improved	Precision	Wilson CI	Object-cluster CI
SAM3	88	52	40	76.9	[63.9,86.3]	[65.3,87.5]
SAM2.1	181	151	134	88.7	[82.7,92.9]	[83.4,93.5]
SeC	163	130	119	91.5	[85.5,95.2]	[86.6,95.9]

and SeC, without retraining. These additional populations support transfer but do not remove the uncertainty of the SAM3 estimate.

Seen and unseen LVOS objects. An audit against matched SAM3-only trajectories covers all 239 objects in 140 validation videos. Table 23 separates untouched objects from those modified by the controller. Eight of 202 seen objects regress, a harmful-intervention rate of 4.0% over that split. The harmful cases concentrate in crowded scenes with multiple similar instances. Among the six modified unseen objects, five improve. Thus, the split disparity is associated with a small, different distribution of intervention outcomes, rather than a broad degradation of seen-object tracking.

Table 23: Object-level intervention outcomes on LVOS-v2. Improved/unchanged/regressed counts are among modified objects, not counts of individual commits.

Split	Objects	Untouched	Improved	Unchanged	Regressed
Seen	202	185	7	2	8
Unseen	37	31	5	0	1

Failure modes on already-accurate predictions. The already-good cell in Table 1 is a corrector-only stress test: recovery is applied unconditionally to 551 samples with mean SAM3 IoU 0.955. It does not use the deployed verifier or selective commit policy. The median IoU change is 0.00, and 91.1% of examples remain within ± 0.05. The negative mean (-0.043) is driven mainly by 26 failures—17 incorrect absence predictions and nine distractor captures—accounting for 86.6% of the total loss. This is not explained by small coordinate perturbations alone. It motivates selective intervention while also showing that acceptance gates do not guarantee harmless correction.

I Inference Cost and Online Operation

Table 24 reports end-to-end measurements on all 160 SecVOS videos using a single H100. Time and compute are benchmark totals divided by 160, not peak latency at an intervention. The 8B verifier and 4B corrector incur substantial overhead over SAM3. FP8 verification reduces mean runtime with a small score decrease, and doubling the occlusion polling interval from 12 to 24 frames reduces it further. These are measured accuracy–cost operating points, not a guarantee of real-time performance or bounded response latency.

Each agent query uses its current frame and preceding visual observations. Temporal verification can defer intervention while later query frames arrive, and correction can require additional model calls. Online operation therefore does not imply zero decision delay. The offline VLM-tracking diagnostic in Appendix K is explicitly different: it receives frames sampled from the complete sequence.

J Trigger Calibration and Dense Verification

Trigger settings were selected on the held-out SA-V training shard `sav_011`: 384 sequences, 1,407 object tracks, and 11,664 ground-truth failure frames. This shard is disjoint from corrector training and the evaluation benchmarks. Among the evaluated candidate settings, the selected configuration was the lowest-cost Pareto-optimal point, requiring 2.27 verifier calls per object; no tested alternative detected more failures at equal or lower cost. This claim concerns the evaluated candidates, not global optimality of the trigger policy.

Table 24: Measured inference cost on full SecVOS-160, single H100. Time and compute are per video (benchmark total divided by 160).

Configuration	Time / video	Relative	Compute	$\mathcal{J}\&\mathcal{F}$
SAM3 (frozen propagator)	12.2 s	1.0×	1.378 PFLOP	74.21
Dual agent, bf16 (canonical)	71.8 s	5.9×	2.22 PFLOP	**76.9**
Dual agent, FP8 verifier	56.2 s	4.6×	2.22 PFLOP	76.8
Dual agent, FP8, occlusion poll every 24 frames	47.7 s	3.9×	2.01 PFLOP	76.41

Table 25 evaluates dense verification on a randomly selected 20-sequence SecVOS subset. The dense configuration removes the confidence guard and invokes verification at every active track frame. It therefore changes both invocation density and guard handling. Its higher score requires 23.4 times as many verifier calls and 12.7 times as many commits as the deployed schedule. The small subset supports an accuracy–cost diagnostic, not a full-benchmark conclusion or an isolated effect of invocation frequency.

Table 25: Dense-verification diagnostic on the same 20 SecVOS sequences. Scores are percentages.

Configuration	Verifier calls	Commits	$\mathcal{J}\&\mathcal{F}$	$\mathcal{J}$	$\mathcal{F}$
SAM3	0	0	73.0	72.8	73.1
Dual agent	168	13	75.2	75.1	75.3
Every active frame, guard removed	3930	165	77.2	77.7	76.8

K VLM as a Standalone Tracker

On 20 SecVOS videos, Qwen3-VL-8B-Thinking receives the initial object annotation and 64 uniformly sampled frames from the entire sequence as native video input. In one call per object, it predicts one box for each sampled frame. SAM3 is initialized on the first frame and propagates through those same 64 frames. Both methods are evaluated using box IoU and AP@0.5 against boxes enclosing the ground-truth masks. This is an offline diagnostic with a matched frame budget, not an online baseline with identical temporal access.

Table 26: Standalone tracking with a matched 64-frame budget on 20 SecVOS videos.

Tracker	Mean box IoU	AP@0.5
Qwen3-VL-8B-Thinking	0.120	0.069
SAM3	0.804	0.827

SAM3 outperforms the VLM on all 20 videos. The result supports retaining the propagator for dense temporal correspondence and allocating VLM calls to verification and recovery.

L Single-Object Tracking Protocols

Table 7 uses native box initialization on LaSOT-ext (150 sequences), TNL2K (700 sequences), and VOT-LT2022 (50 sequences). Each sequence runs in a separate process. LaSOT-ext and TNL2K metrics are checked against the official pysot implementation; VOT-LT2022 uses the official long-term analysis in the VOT toolkit, without re-initialization. We report success AUC, precision at 20 pixels, and normalized precision at threshold 0.2 for the first two datasets. VOT precision, recall, and F-score follow its toolkit convention. TNL2K uses bounding-box initialization without a language prompt.

The benchmark definitions follow LaSOT-ext [10], TNL2K [22], and VOT2022 [16]. Table 27 gives the complete SAM3 and dual-agent metrics corresponding to the main-paper comparison.

The agents use the original checkpoint-9418 recovery model and verifier. TNL2K uses the plain tracking configuration. The VOT-LT run additionally uses the propagator-confidence guard, and the

Table 27: Complete box-initialized tracking metrics. LaSOT-ext and TNL2K scores are percentages; P_{norm} is normalized precision at 0.2. †Size-selected composite.

Benchmark	Metric	SAM3	Dual agent
LaSOT-ext	AUC	63.70	64.29†
	Precision	74.77	75.46†
	P_{norm}	72.86	73.54†
TNL2K	AUC	65.20	66.50
	Precision	72.75	74.13
	P_{norm}	74.25	75.70
VOT-LT2022	Precision	0.6318	0.6104
	Recall	0.5735	0.6130
	F-score	0.6012	0.6117

LaSOT-ext result is a size-selected composite: recovery results are used for large-object sequences and SAM3 results for small-object sequences (threshold 60 pixels on the sequence-level median object side). This composite is not evidence for an unrestricted online size-selection policy. The unselected full-benchmark LaSOT-ext recovery run was not measured. These adaptations must be considered when interpreting transfer from the VOS configuration.

The verified box-initialized scores differ from the mask-initialized runs reported in the rebuttal. Table 28 preserves both. The rebuttal reports normalized-precision AUC, whereas the main table reports normalized precision at 0.2, matching the literature comparison. Those quantities are not interchangeable. The TNL2K box-initialized AUC gain is 1.30 points with a paired sequence-bootstrap 95% interval of $[0.47, 2.19]$; this interval describes evaluation-sequence uncertainty, not variation across training runs.

Table 28: Initialization-specific comparisons. LaSOT-ext reports the size-selected composite. AUC scores are percentages; VOT-LT2022 reports F-score. Each comparison uses the matching initialization for both methods.

Benchmark	Initialization	SAM3	Dual agent
LaSOT-ext (AUC)	Mask	60.85	61.13
	Native box	63.70	64.29
TNL2K (AUC)	Mask	62.74	63.94
	Native box	65.20	66.50
VOT-LT2022 (F)	Mask	0.590	0.601
	Native box	0.601	0.612

M Scope and Limitations

The added cost is measured in Appendix I, and the current implementation is not real-time on a single GPU. Average runtime does not characterize worst-case intervention latency. Incorrect verification or correction can override an accurate track and affect subsequent memory, as quantified in Appendix H; the gates reduce opportunities for harm but provide no formal safety guarantee.

Transfer experiments cover several propagators and natural-video benchmarks. They do not establish performance in specialized domains such as endoscopic imagery, where additional failure mining or model adaptation may be needed. We cannot verify video-level overlap between the foundation models' pretraining data and the evaluation benchmarks. The standalone-VLM diagnostic does not resolve that uncertainty. Finally, confidence intervals for measurable commits reflect limited and selectively annotated populations, and do not quantify the full variation across model training seeds.

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